

GAURAV KUMAR

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Education

Guru Tegh Bahadur Institute of Technology, GGSIPU
Bachelor of Technology in Information Technology — CGPA: 8.3/10

New Delhi, India
Sep. 2024 – July 2028

Technical Skills

Languages: Java, JavaScript, SQL
Frameworks & Libraries: React.js, Node.js, Express.js, TailwindCSS
Databases: MongoDB, MySQL
Developer Tools: Git, GitHub, Postman

Projects

Java-Blockchain – UTXO Blockchain Engine | *Java, BouncyCastle (ECDSA), GSON, SHA-256* [GitHub](#)

- Built a **Bitcoin-style UTXO blockchain** in Java with full transaction lifecycle – coinbase issuance, fee-prioritized mempool, double-spend prevention, and **satoshi-integer arithmetic** to eliminate floating-point precision errors.
- Implemented **ECDSA wallets** (BouncyCastle `prime192v1`) with key-pair generation, per-transaction signing & verification; built a **fee market** where miners earn block reward plus fees, prioritizing highest-fee mempool transactions.
- Built a **Proof-of-Work engine** with nonce-based SHA-256 solving, **Merkle root** integrity in block headers, and **dynamic difficulty retargeting** every N blocks based on actual vs. target mine time.
- Extended to a **2-node P2P network** via Java Sockets with 4-rule consensus validation (hash, PoW, ECDSA, UTXO) before appending, refactored into a **4-layer architecture** separating state, mining, validation, and orchestration.

FileShare – Secure File Sharing Platform | *React.js, Node.js, Express.js, MongoDB, Cloudinary* [GitHub](#)

- Built a full-stack file sharing platform supporting guest and authenticated users with role-based feature access and a clean, mobile-responsive React 19 interface.
- Implemented **advanced security**: JWT authentication, Bcrypt.js password hashing, per-file password protection, and custom expiry dates; guest uploads auto-expire after 24 hours.
- Integrated Cloudinary for cloud storage (50MB/file, 10 files/upload), QR code generation, short URL sharing, Nodemailer email sharing, and per-file download analytics.
- Designed 15+ RESTful API endpoints covering upload, search, pagination, and password verification flows for both guest and registered users.

Yogya-Setu – Internship Management Platform | *React.js, Node.js, Express.js, MongoDB, JWT* [GitHub](#)

- Built a full-stack internship platform with **ML-powered matching** using **MiniLM-L12 semantic embeddings** to recommend relevant internships based on parsed resume profiles.
- Designed a **3-service microservices architecture**: React frontend, Node.js/Express REST API with JWT & OTP auth, and a Python ML service for semantic similarity scoring.
- Implemented profile management with **completeness scoring**, resume parsing, application tracking, AI-generated quizzes, and multi-language (i18n) support.
- Built RESTful APIs for internship search, recommendation ranking, and application workflows with pagination, filtering, and automated ML-based match scoring.

Certifications

- **Smart India Hackathon Winner – 2025**
- **Postman Api Fundamentals Student Expert** – Postman
- **Introduction to the Internet of Things and Embedded System** – Coursera

Achievements

- **Smart India Hackathon (SIH) 2025 – National Winner** – Awarded 1st place in the Government of India's flagship national-level hackathon, competing against 1 lakh+ participants across India.
- **DSA – 200+ Problems Solved across different platform** – LeetCode, GeeksforGeeks, and CodeChef, Hackerrank